



Heart Disease Prediction

using Machine Learning

Group # 3 | DSAI Department

Savaira Majeed — DSAI231103031

Faizan Nazik — DSAI231103023

Shaesta Saleem — DSAI231103043

Sarfraz Ahmad — DSAI231103048

Mustajab Zahra — DSAI231103016

Muhammad Gulfam — DSAI231103047

Instructor: Ms. Allah Rakhi Saman

Project Overview

Objective

Predict the presence of heart disease in patients using multiple classification algorithms, compare their performance, and identify the most accurate model.

1

Logistic Regression

Baseline linear model for binary classification

2

Random Forest

Ensemble of decision trees for improved accuracy

3

XGBoost

Advanced gradient boosting — best performer

Step 2: Dataset Overview

920

Total Patients

16

Features

509

Heart Disease +ve

411

Heart Disease -ve

Key Features Used

age

Patient age

chol

Serum cholesterol

sex

Gender (Male/Female)

thalch

Max heart rate achieved

cp

Chest pain type

oldpeak

ST depression

trestbps

Resting blood pressure

ca

major vessels

Target Variable: num → converted to binary (0 = No Disease, 1 = Disease)

Step 3: Data Exploration & Visualization

Disease Distribution

Class 0 (No Disease): 411 patients
Class 1–4 (Disease): 509 patients
→ Slight imbalance towards disease positive

Age Distribution

Patients range from 28–77 years
Peak age group: 50–60 years
Normal-like bell-curve distribution

Gender Distribution

Male patients: ~720 (majority)
Female patients: ~200
Males have higher representation

Correlation Heatmap

Positive correlation: age ↔ trestbps
Negative correlation: thalch ↔ num
CA and num show strong positive link

Step 4: Data Cleaning & Preprocessing

Missing Values Handled

slope: 309 missing

ca: 611 missing

thal: 486 missing

fbs: 90 missing

treanv: 10 missing

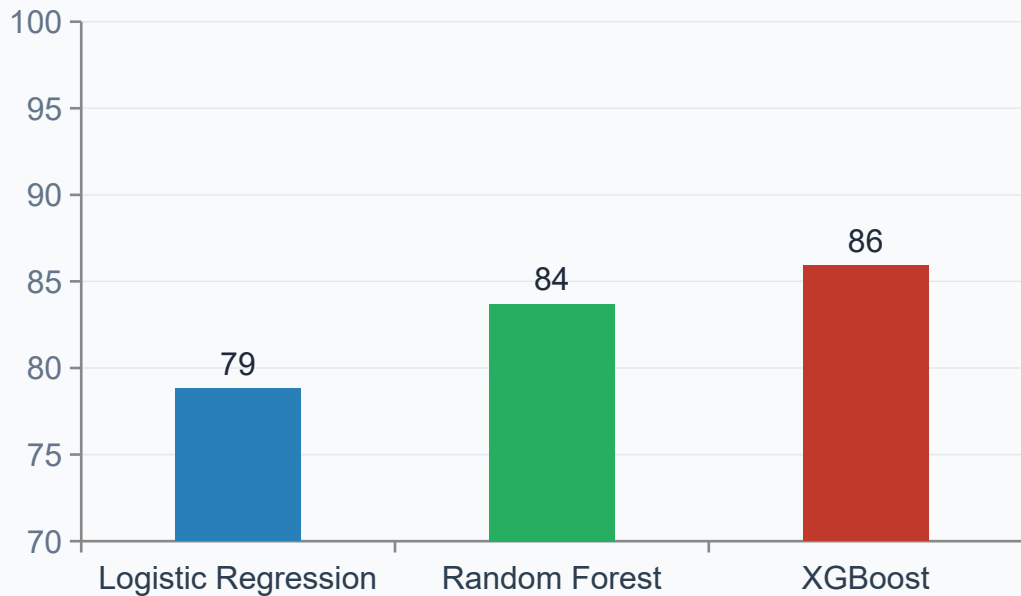
Preprocessing Steps

- Numerical columns → filled with median
- Categorical columns → filled with mode
- Dropped: id, dataset, num columns
- Target: num > 0 → binary (0 or 1)
- Label encoding for all categorical features
- 80/20 Train-Test split (random_state=42)

✓ Result: Training Set — 736 samples (13 features) | Testing Set — 184 samples (13 features)

Step 6: Model Training & Results

Model Accuracy Comparison (%)



Logistic Regression

Accuracy: 78.80%

Random Forest

Accuracy: 83.70%

XGBoost ★ Best

Accuracy: 85.87%



XGBoost outperforms all other models with 85.87% accuracy

Step 7: XGBoost Model Evaluation

Confusion Matrix

	Predicted: No	Predicted: Yes
Actual: No	67 TN	8 FP
Actual: Yes	18 FN	91 TP

Classification Report (XGBoost)

Class	Precision	Recall	F1-Score
0 (No Disease)	0.79	0.89	0.84
1 (Disease)	0.92	0.83	0.88
Accuracy			0.86
Macro Avg	0.85	0.86	0.86
Weighted Avg	0.87	0.86	0.86

High Precision (0.92) for disease class reduces false alarms — critical in medical diagnosis

Confusion Matrix Comparison — All Models

Logistic Regression

62	13
TN	FP

26	83
FN	TP

Accuracy: 78.80%

Random Forest

62	13
TN	FP

17	92
FN	TP

Accuracy: 83.70%

XGBoost ★

67	8
TN	FP

18	91
FN	TP

Accuracy: 85.87%

XGBoost achieves the fewest False Positives (8) — meaning fewer healthy patients are misdiagnosed

Step 9: Streamlit GUI — Heart Disease Predictor



An interactive web application built with Streamlit for real-time heart disease prediction using the trained XGBoost model.



Patient Input Form

Users enter patient details: age, sex, chest pain type, blood pressure, cholesterol, heart rate, and more via sliders and dropdowns.



Instant Prediction

XGBoost model runs in real-time. Results displayed instantly as 'Heart Disease Detected' or 'Patient is Healthy' with color-coded alerts.



Visual Feedback

Shows prediction confidence, input summary table, and feature importance charts for model interpretability.



Saved Model

Uses the pre-trained XGBoost model (heart_disease_model.pkl) loaded with joblib for fast inference without retraining.



Conclusion

🏆 XGBoost achieved the highest accuracy of 85.87%, outperforming Logistic Regression (78.8%) and Random Forest (83.7%)

🎯 The model achieves Precision of 0.92 for the disease class — minimizing dangerous false negatives in clinical settings

🔄 Real-world missing data was handled effectively using median/mode imputation and label encoding for all 13 features

💻 The Streamlit GUI enables non-technical users to interact with the model in real-time for practical clinical use

Thank You!